Geometry masks and track-field annotation

LHCb Track Acceptance

One set of tracks. Six detector setups. Both magnet polarities.

Transport through the magnetic field, test the detector surfaces, and return labelled hit counts and acceptance flags.

User guide and default output contract

19 October 2026

What the tool does

1. **Load** saved detector XY masks at their global z positions, plus the Bx/By/Bz field map.
2. **Propagate** each input track forward through the field. With both polarities, run two trajectories, not one per geometry.
3. **Count** logical detector layers intersected by each trajectory. Several sensor surfaces can contribute to one counting layer.
4. **Annotate** the original track rows with integer counts and boolean cuts, labelled by geometry and polarity.

Output: NumPy arrays in input-track order. No event selection, decay reconstruction or ntuple rewriting is performed automatically.

Choose the interface that matches the study

Interface	Geometry choices	Field choices
`DetectorTransport`	One saved setup	One polarity
`GeometryComparison`	Six defaults, or a custom mapping	One polarity
`PolarityComparison`	Six defaults, or a custom mapping	Both polarities

`PolarityComparison.label_tracks(states)` is the recommended reusable interface for comparing the default layouts.

For a single call, use `label_tracks_both_polarities(...)`. For large samples, keep the service loaded and process bounded chunks.

The six default geometries

UP borders	Reduced FT	Baseline FT
`gdml`: nominal	8 / 8 / 10	10 / 10 / 12
`y8`: top/bottom cutoff	8 / 8 / 10	10 / 10 / 12
`y8x6`: also left/right cutoff	8 / 8 / 10	10 / 10 / 12

All six use FT_withHole. The numbers specify module widths in T1 / T2 / T3, not numbers of counting layers.

Each setup includes TV, UP, MP, FT, RICH1, RICH2, PicoCal and Muon M2–M5.

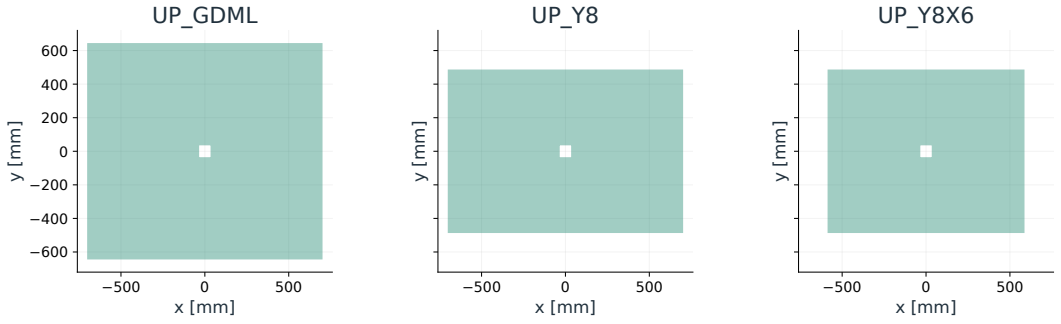
MagUp and MagDown give **12 geometry/polarity predictions per track**.

Default detector inventory

System	Surfaces	Counting layers	Global z [mm]
TV	100	50	-288.0–558.1
RICH1	2	2	795.0–2237.2
UP	16	4	2314.0–2656.0
MP	5	5	7665.5–9132.5
FT	12	12	7902.7–9384.7
RICH2	2	2	9487.5–11839.2
PicoCal	2	2	12565.9–12601.0
MUON	4	4	15270.0–18870.0

A layer fires when *any member surface* contains the track crossing at that surface's own z. Surfaces upstream of the starting state do not fire.

UP: three border choices, one counting convention



Outer: 1393.7 x 1279.8 mm
Central hole envelope: 77.3 x 79.9 mm

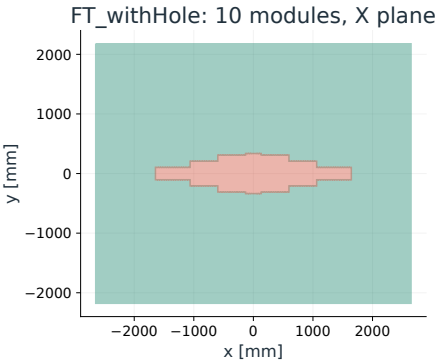
Outer: 1393.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm

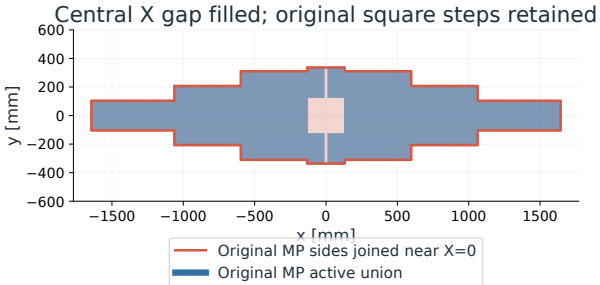
Outer: 1164.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm

The source masks preserve their central hole and sensor pattern. Each variant retains 16 surfaces grouped into four logical layers.

`nUP` is 0–4. `hasUP` means $n_{\mathrm{UP}} > 0$; it does **not** mean all four layers were crossed.

FT_withHole and the MP region





The FT hole is the MP left/right envelope joined across its central x gap. Square outer steps are preserved; there is no convex interpolation.

The 3288.4 $\times$ 672.6 mm envelope is subtracted in global XY from every FT plane, after stereo rotation. It is not a projection between detector z positions.

FT and MP hit counts are the main comparison

FT

T1, T2, T3 each contain X/U/V/X at $0, -5, +5, 0$ degrees.

Both module layouts have **12 counting layers**.

Module width: 529.3 mm. The 8, 10 and 12 module X widths are 4234.4, 5293.0 and 6351.6 mm.

MP

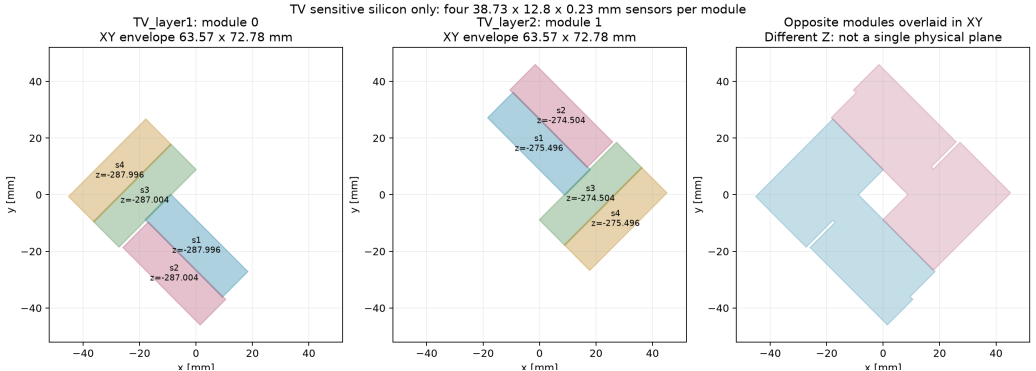
The granular sensor acceptance is reduced to five logical XY unions.

`nMP` is 0–5; `nFT` is 0–12.

MP geometry is unchanged between FT choices. With identical trajectories, n_{MP} should match track by track.

Keep both counts to study their joint coverage and to change cuts offline.

TV: real silicon positions, one hit per module



200 sensitive-silicon sensors form 50 modules, represented by 100 sensor-centre planes. Left/right acceptances and sensor gaps are retained.

Sensor centres span −287.996 to 558.146 mm. From the origin, 29 modules are downstream and 21 cannot fire. The earlier 20-plane square mockup is not used.

TV and MT cuts returned by default

Property	Exact requirement
`hasTV_3Hits`	$n_{\mathrm{TV}} \geq 3$ counted modules
`hasTV_4Hits`	$n_{\mathrm{TV}} \geq 4$ counted modules
`hasMT`	$(n_{\mathrm{MP}} \geq 3) \vee (n_{\mathrm{FT}} \geq 10) \vee [(n_{\mathrm{MP}} \geq 1) \wedge (n_{\mathrm{FT}} \geq 6)]$

MT thresholds are configurable using `MTSelection`, globally or separately for each geometry. Changing a cut does not change the masks.

TV and UP are **not included** in `hasMT`. Apply the analysis's TV/UP requirements separately. Keep the integer counts for later cut scans.

How to read an output name

`hasMT_up_y8x6_ft_8_8_10_withHole_magup`

Part	Meaning
`hasMT`	Property or count name
`up_y8x6`	UP border choice
`ft_8_8_10_withHole`	FT widths and MP-derived hole
`magup`	Field-polarity prediction

41 properties $\times$ 6 geometries $\times$ 2 polarities = 492 columns.

All arrays have one value per input track. Per-layer `fired_*` columns are omitted unless `include_layer_hits=True`.

Counts and detector-presence flags

Count	Range	Presence flag	Interpretation
`nTV`	0–50	`hasTV`	Fired modules; from the origin at most 29 reachable
`nUP`	0–4	`hasUP`	Four logical UP layers, not 16 independent counts
`nMP`	0–5	`hasMP`	Five simplified MP layers
`nFT`	0–12	`hasFT`	Four FT layers per station
`nRICH1`	0–2	`hasRICH1`	Radiator entry/exit acceptance planes
`nRICH2`	0–2	`hasRICH2`	Radiator entry/exit acceptance planes
`nPicoCal`	0–2	`hasPicoCal`	Simplified entry/exit surfaces
`nMUON`	0–4	`hasMUON`	Stations M2–M5

Every presence flag means **count** > 0, not full detector coverage. Use `nUP == 4` or `nFT == 12` when all logical layers are required.

RICH, PicoCal and Muon: useful canonical flags

Flag	Meaning
`hasRich1Entry`, `hasRich1Exit`	RICH1 plane hits
`hasRich2Entry`, `hasRich2Exit`	RICH2 plane hits
`hasRichEntry`, `hasRichExit`	OR of the corresponding RICH1/RICH2 hits
`hasPicoCal`	At least one PicoCal surface hit
`inPicoCal`	Both PicoCal entry and exit hit in the default layout
`nMuon`, `hasMuon`	Aliases for `nMUON`, `hasMUON`
`isMuon`	At least 3 Muon stations; adjustable with `min_muon_stations`

These are **geometric labels**: no Cherenkov PID, calorimeter response or reconstructed muon identification is simulated.

Additional volume flags and complete output reference

For each of `RICH1`, `RICH2` and `PicoCal`, the API also returns:

Property pattern	Meaning
`<system>_entry`	Entry plane fires
`<system>_exit`	Exit plane fires
`<system>_entry_exit`	Entry AND exit fire
`<system>_entry_only`	Entry fires AND exit does not

The same geometry and polarity suffix is appended to *every* property.

`tool_output_schema.json` lists all 492 exact column names, their NumPy dtypes and values for two runnable example tracks.

Minimal runnable example

```
from xyshapes_detectors import PolarityComparison

states = dict(
    x=[0., 0.], y=[0., 0.], z=[0., 0.],
    tx=[.04, .02], ty=[.01, -.01],
    q_over_p=[1/20000., -1/50000.],
)
service = PolarityComparison.from_files(
    "MagField/magnetic_field_v5r0.root",
    map_polarity="magdown",
)
new_fields = service.label_tracks(states)
```

Positions are mm; q/p is $1/\mathrm{MeV}$. Input slopes are p_x/p_z and p_y/p_z. Use each charged daughter's production point; $p_z > 0$ is assumed.

Example output: two tracks, one prediction

`up_y8x6_ft_8_8_10_withHole_magup`

Track	nTV	nUP	nMP	nFT	hasMT	hasPicoCal
1	7	4	3	0	true	false
2	4	4	5	0	true	true

Values are generated by running the preceding example against the bundled field map, not hand-entered illustrations.

The same tracks also receive the other 11 geometry/polarity predictions. The service does not filter, reorder or drop the input rows.

Both polarities are predictions for the same tracks

- The bundled map is labelled MagDown. The MagUp prediction reverses **Bx, By and Bz together**; another map can declare its own reference polarity.
- Each polarity requires a separate transport. Within that transport, all six geometries reuse the same physical-plane crossings.
- Keep both `_magup` and `_magdown` columns, then select the appropriate ones using the data sample's actual magnet metadata.
- Opposite-field flags do **not** measure the event's magnet polarity. The two fields are ideal sign reversals, not separately measured maps.

The same-sign comparisons isolate geometry changes; the same-geometry comparisons isolate the field-polarity dependence.

Attach fields to daughters without losing decay alignment

```
output = dict(original_columns)
for daughter, states in daughter_state_columns.items():
    flags = service.label_tracks(states)
    output.update({f"{daughter}_{name}": values
                   for name, values in flags.items()})
```

- Keep one row per candidate for fixed-daughter trees; prefix by daughter.
- For jagged collections, restore track offsets after flattening.
- Combine daughter flags into decay cuts in the analysis code.
- Write a new output file or friend tree; ntuple I/O is not automatic.

Scope and checks

Included

Field-map interpolation, forward RK4 transport, XY acceptance at actual surface z, logical-layer counts and configurable geometric cuts.

118 tests passed: field/charge reversal, grouping, upstream masks, MT thresholds, two-polarity consistency and track order.

Not simulated

Material scattering, energy loss, sensor efficiencies, reconstruction, detector response or PID.

Failed/curling or incomplete transport raises `TrajectoryError`. It must not be silently converted into an acceptance miss.

Toy geometric acceptance is not a physics-sample efficiency. Use the sample's track distribution, weights and denominator when comparing decay yields.

Start from a clone

```
git clone --branch feature/detector-mask-loader \
  https://github.com/qgl90/xyshapes_detetctors.git
cd xyshapes_detetctors
pip install -e '.[field]'
python -m unittest discover -s tests
```

The geometry masks and ROOT field map are included. ROOT itself is not required for mask loading or propagation; uproot reads the field histograms.

Usage: `docs/decay_study_quickstart.md`
API: `docs/geometry_comparison.md`
Exact outputs: `docs/geometry_documentation/tool_output_schema.json`
Geometry and TV studies: `acceptance_study_summary.pdf`